IAT 461 Assignment 3 Report

A1. Data acquisition and cleaning

Missing values (coordinates, employees, feepaid)

I looked at why each column is missing first, cause that tells me how to fix it. I am not investigating every missing column, only the ones that feed a feature or a filter. The address columns (unittype, unit, house, street), the trade name and near-constant fields like province and country I leave alone, since none of them feed a feature and the README says not to hand-fix messy fields.

feepaid. I filter. The 37% missing is not random, it is missing by year: 84.0% for FY24 but only 0.7% and 0.6% for FY25 and FY26. So it is structural, the city just didn't publish FY24 fees. If I filled the blanks with a guess I would be making up fake fees for about 47k rows, which would feed straight into my A3 Size features. Instead I just drop FY24 later and the problem goes away on its own.

numberofemployees, I leave it alone. 35.8% of rows are 0, which looks like missing data, but the zeros are almost all Long-term Rental and Short-term Rental Operator. Those are properties, they really do have 0 staff, so a 0 is a real number here, not missing. Imputing it would erase a real signal, so I don't touch it.

coordinates, I drop those rows. Only 50.3% of rows have lat/lon, there is no way to recover a missing coordinate and A2 clusters on location only, so a row without a coordinate can't take part. It is a big cut (about half the data) so I state it explicitly rather than filtering it quietly.

Filtering to a sensible subset

I filter in three steps and print how many rows survive each one. status Issued (167008 rows): the other statuses are Pending, Gone Out of Business, Inactive and Cancelled, which are licences that aren't active businesses, so including them would mean clustering businesses that don't really exist. FY26 (56942 rows): this fixes feepaid (drops to 0.6% missing so I never impute), avoids counting the same business across multiple years and keeps the sample small enough to plot and re-cluster fast. has coordinates (29025 rows): the biggest single cut, from 56942 down to 29025, cause a missing coordinate can't be recovered.

Two columns I checked but don't fix are businesssubtype (89.5% missing), so little real information that I don't rely on it and postalcode (46.6% missing), so if I need a location grouping I'd rather use localarea, which is far more complete.

Consolidating businesstype

The cleaned data has 89 business types and most of them are tiny, so I keep the biggest ones and group the rest into one Other bucket. To pick how many, I looked at cumulative coverage: top 10 covers 63.4%, top 15 covers 74.0%, top 20 covers 80.8% and top 25 covers 85.0%. Each extra jump adds less (10 to 15 adds about 10 points, 15 to 20 adds about 7, 20 to 25 adds about 4), which is the long tail flattening out. The README suggests covering about 80 to 90% and top 20 is the smallest cut that lands in that range at 80.8%, so I keep the top 20 plus Other (21 categories total). Other holds 19.2% of rows and sits in the middle of the pack, which tells me I didn't cut too hard.

Refinement. I also combined businesstype and businesssubtype into a finer industry label where a subtype exists, falling back to the plain type otherwise. This takes 89 categories to 145 (56 more) and the clearest example is Restaurant, which splits into Restaurant - Class 1 with liquor service and Restaurant - Class 1 no liquor service. But only 19.4% of rows have a subtype, so 80% look the same as before and more categories means a sparser feature space. So the finer label is sharper for a fifth of the data but costs more sparsity and I keep businesstype (the top 20 version) as my main industry feature since it is cleaner.

Final cleaned dataset: 29025 licences, with complete coordinates, complete localarea, near-complete feepaid (99.2%) and a tidy 21-category business type column.

A2. Location-only clustering: Kmeans vs. DBSCAN

Choosing K with the elbow method

I ran Kmeans for K from 1 to 10 and looked at how much each extra cluster reduces inertia (the drop column). The first two drops are big (19.6 then 8.3), so going from 1 to 3 clusters captures real structure. Going from 3 to 4 still helps a decent amount (4.8), but after 4 the drops get small (2.3 then 1.5) and stay small and about the same. Once the drops flatten out I've gone past the elbow, so I use K4. One thing I noticed is that the elbow is soft, a gentle curve rather than a sharp corner, which I think reflects that businesses are a continuous spread that is just denser in some spots rather than a few clean separate blobs.

How the Kmeans clusters map onto Vancouver geography

The 4 clusters line up with real parts of the city. The big red cluster in the middle is the busy downtown and it makes sense it has the most businesses (16177, more than half) since that is the center of the city. The cyan cluster on the left is the west side, the pink cluster at the bottom is the south and the blue cluster on the right is the east side. So the four groups match areas I'd recognize (center, west, south, east). The downtown cluster grabbing over half the businesses shows the city isn't spread evenly, the center is far busier than the edges.

How DBSCAN compares and what noise means

DBSCAN doesn't take a number of clusters, it groups by density using eps (how close businesses must be to count as neighbors) and min_samples (how many neighbors make a dense area) and it labels sparse leftover points as noise. I measured distance in real kilometres using the haversine metric so eps is meaningful. A quick sweep showed that as eps grows the clusters merge (41 down to 3) and noise stays very low everywhere (never above 7%), which is itself a finding: businesses are densely packed almost everywhere in the city, so barely anything is stranded as an outlier. I settled on eps = 0.4 km and min_samples = 30, giving 11 clusters and 0.8% noise, a readable middle ground.

The two methods tell a different story. Kmeans made 4 even regions and forced every business into one. DBSCAN made one giant cluster covering the whole dense middle of the city, cause that area is all connected with no real gap to split on, plus a few small pockets that are genuinely separated. So both agree the center is packed, but Kmeans draws a line straight through it while DBSCAN refuses to split it. This comes down to assumptions: Kmeans assumes round even blobs and assigns everyone, so it always returns exactly 4 regions, while DBSCAN assumes dense areas separated by empty space, so it only splits at real gaps and is allowed to leave points out. The 239 noise points (0.8%) are businesses sitting alone in sparse spots without enough neighbors to join a cluster; Kmeans had to fold those same lonely dots into a region anyway, but DBSCAN marks them as noise.

A3. Feature-based clustering: Size, Industry, Lifecycle

Features and scaling

Size uses number of employees and fee paid. Both are very skewed (median 3 employees but a max of 4000; median fee 277 but a max of 63722), so a few giants would take over the clustering. I log-transformed both to compress the giant tail and spread out the crowded low end. Lifecycle uses licence duration (expire minus issue, median 377 days, about a year), the month issued encoded cyclically with sin and cos so Dec and Jan sit next to each other instead of looking far apart and the revision number. Industry uses one-hot encoding of the top-20 businesstype so no category is treated as bigger than another, giving 21 columns.

I combined all three groups into one 27-column table and scaled with StandardScaler. Scaling is essential here (unlike A2) cause the features are in wildly different units: duration is in the hundreds while the industry columns are just 0 or 1, so without scaling duration would dominate the distance and the industry flags would barely count. After scaling the mean is 0 and the spread is 1, so every feature has a fair say.

What the clusters represent

The elbow for the feature clustering is soft (the drops stay around 28k to 37k the whole way down rather than collapsing), which reflects that businesses vary smoothly rather than falling into clean groups, so I used K=4 where the biggest drops stop. PCA to 2D captures only 15.8% of the variation (cause the 21 sparse industry columns spread it thin), but the four clusters still separate into their own regions, so Kmeans found real structure.

Profiling each cluster by its average size, lifecycle and top industries gives clear identities. Cluster 0 (1594) is restaurants, literally 100% restaurants, pulled out purely by industry. Cluster 1 (3640) is property businesses, mostly Long-term Rental plus Parking, with the highest fees and lowest employees, which fits since rentals and parking pay a fee but have no staff. Cluster 2 (4990) is defined by lifecycle rather than industry, with very short licence durations (below average) and more revisions, across mixed industries, so it is a short or amended licence group. Cluster 3 (18801) is the big normal group, everything near average with a slightly longer full-year licence and a broad industry mix, the typical everyday business.

This is very different from A2. In A2 I clustered on location, so the groups were places (downtown, west, south, east). Here I clustered on what a business is (size, lifecycle, industry), so the groups are business types (restaurants, rentals, short-licence businesses). A2 tells me where businesses are; A3 tells me what kind they are. Two businesses next to each other downtown share an A2 cluster, but if one is a restaurant and the other a rental they land in different A3 clusters, so the two clusterings answer completely different questions about the same businesses.

B4. Cluster membership and interpretation

At K=4 the app grouped the 22 neighborhoods as follows. Cluster 0 is Kitsilano, West End, South Cambie and Shaughnessy, grouped by very high Long-term Rental (28%) which fits since these are dense areas with lots of apartments. Cluster 1 is the big group of 12 areas (Fairview, Kerrisdale, Oakridge, Killarney, Dunbar and others), with Health Care as its top type (22%), the quiet spread-out residential neighborhoods where the mix is everyday clinics and shops. Cluster 2 is Mount Pleasant, Grandview-Woodland, Strathcona, Marpole and Sunset, a middle group with a moderate mix and nothing extreme. Cluster 3 is Downtown alone, whose top types are Legal Services (16%) and Business Support Services, the office and business core, so it doesn't look like anywhere else.

Does the grouping make sense?

Yes. The groups match what I'd expect from knowing the city, especially Downtown standing on its own as the legal and business core. That tells me the composition matrix picked up something real rather than random.

Anything surprising?

Shaughnessy ended up with Kitsilano, which is a little surprising cause they feel different, Shaughnessy is a quiet upscale area while Kitsilano is busy and young. But both have a lot of Long-term Rental licences, so the app grouped them on that shared feature. The takeaway is that areas that feel different in real life can still match if their business mix lines up.

B5. Reflections

Do business-similar areas tend to be geographically close, or does similarity cut across the map?

Mostly it cuts across geography. On the map the areas in the same cluster aren't sitting together in one spot for example, the rental group (Kitsilano, West End, South Cambie, Shaughnessy) is spread across the west and center and the big health-care group is scattered all over. So the grouping is driven by business mix, not location and two neighborhoods on opposite sides of the city can share a cluster (I mentioned this finding in the B4)

How does this compare to the Part A Industry clustering?

It is a partly consistent story. In A3 the business clusters were things like rentals, restaurants and a big normal group and in Part B the areas also split partly on rentals and health care. So rentals show up as a strong distinguishing feature in both parts, which is a nice consistent thread, but they aren't identical: A3 grouped individual businesses by what they are, while Part B groups whole areas by their mix, so it is a related story at a different zoom level.

Pick one area you didn't expect to see grouped with another — what do they have in common?

One surprising grouping. Shaughnessy. I made some search about Shaughnessy and found out it is a quiet fancy neighbouhood area so it's lowkey interesting to see it is placed together with Kitsilano (very busy and young with chill beach which I love). But both have a lot of Long-term Rental licences, so the app grouped them on that shared feature. So that comes to the interesting finding that even areas that feel different can match if their business mix is similar.